

Dae Kwan Ko

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Research Interests

I am a systems biologist developing causal and predictive regulatory genomics approaches to understand how gene regulatory networks integrate microbial and environmental signals to shape plant resilience. I combine functional genetics, time-resolved transcriptomics, single-nucleus multiome profiling, transcription factor–DNA binding maps, and interpretable machine learning and network modeling to identify causal regulatory mechanisms and generate experimentally testable predictions. My research spans crop stress biology, plant–microbe interactions, and cellular stress signaling, with a long-term goal of defining regulatory principles that predict plant responses across genotypes, microbial contexts, and environments.

Professional Appointments

Assistant Professor (Fixed Term) 2022–Present

MSU-DOE Plant Research Laboratory

Department of Plant Biology, Michigan State University

Affiliate, Great Lakes Bioenergy Research Center

- Research: Leads an integrative program combining single-nucleus genomics, transcriptomics, TF–DNA mapping, regulatory-network inference, and functional validation to identify causal regulators of plant stress responses and environmental resilience.

Postdoctoral Research Associate 2017–2022

MSU-DOE Plant Research Laboratory

Great Lakes Bioenergy Research Center, Michigan State University

- Advisor: Dr. Federica Brandizzi
- Research: Investigated regulatory networks controlling the plant unfolded protein response and stress recovery using functional genomics, network modeling, and genetics, while applying systems approaches to cell-wall biogenesis and differentiation in bioenergy crops.

Postdoctoral Research Associate 2016–2017

Department of Plant Biology, Michigan State University

- Advisor: Dr. C. Robin Buell
- Research: Identified unintended consequences of genome modification methods in clonally propagated diploid potato, addressing critical concerns related to genome editing and its implications within the context of crop improvement.

Education

Ph.D. Plant Biology | The University of Texas at Austin 2009–2016

- Advisor: Dr. Z. Jeffrey Chen
- Thesis: *Clock-Regulatory Networks Contribute to Growth Vigor in Maize Hybrids*
- Research: Circadian clock regulation and heterosis in maize using functional genomics

M.S. Biological Sciences | Seoul National University, Seoul, South Korea. 2006–2008

- Advisor: Dr. Choo Bong Hong
- Thesis: *Submergence-Inducible and Circadian Rhythmic Transcriptional Networks in *Nicotiana tabacum**
- Research: Characterizing clock-regulated transcription factor genes in response to abiotic stress in *Nicotiana tabacum*

B.S. Crop Science | Konkuk University, Seoul, South Korea.

2000–2006

Publications

Peer-Reviewed Articles – 24 publications; 14 as first or co-first author

1. **Ko DK**, Brandizzi F (2026). Predicting transcriptional regulators in plants in the era of artificial intelligence. *Current Opinion in Plant Biology* **93**:102955.
2. **Ko DK**, Brandizzi F (2026). Protocol for applying a network-enabled gene discovery pipeline to non-model plant species. *STAR Protocols* **7**(3):104641.
3. **Ko DK***, Danieli R*, Brandizzi F (2026). Rewiring the unfolded protein response for post-stress growth recovery in plants. *Plant Communications* **7**(7):101924.
*These authors contributed equally.
4. **Ko DK**, Brandizzi F (2026). A tissue-resolved, network-based transcriptomic framework for abiotic stress responses in sorghum. *The Plant Journal* **126**(1):e70834.
 - Highlighted in [GLBRC bulletin](#)
 - Highlighted in [SorghumBase](#)
5. **Ko DK**, Brandizzi F (2025). A network-enabled pipeline for gene discovery and validation in non-model plant species. *Cell Reports Methods* **5**(1):100963.
 - Highlighted in [MSU-PRL bulletin](#)
 - Highlighted in [GLBRC bulletin](#)
6. Adhikari B, Verchot J#, Brandizzi F, **Ko DK#** (2025). Advances in plant UPR: mechanisms, implications, and future directions in viral defense. *Journal of Biological Chemistry* **301**(4):108354.
#Co-corresponding author.
7. **Ko DK**, Brandizzi F (2024). Dynamics of ER stress-induced gene regulation in plants. *Nature Reviews Genetics* **25**(7):513-525.
8. Choi D, **Ko DK**, Kim DH (2024). Transcriptome analysis revealed that Arabidopsis model plant invokes the activation of heat shock proteins and ER stress response against cesium stress. *Plant Biotechnology Reports* **18**(3):385-399.
9. **Ko DK**, Kim JY, Thibault EA, Brandizzi F (2023). An IRE1-proteasome system signalling cohort controls cell fate determination in unresolved proteotoxic stress of the plant endoplasmic reticulum. *Nature Plants* **9**(8):1333-1346.
 - Featured in a Spotlight by Varshney et al., *Trends in Plant Science* (2023)
 - Highlight in [MSUTODAY](#)
 - Author Interview [video](#)
10. Bhandari DD, **Ko DK**, Kim SJ, Nomura K, He SY, Brandizzi F (2023). Defense against phytopathogens relies on efficient anti-microbial protein secretion mediated by the microtubule-binding protein TGNap1. *Nature Communications* **14**(1):6357.
11. **Ko DK**, Brandizzi F (2022). Transcriptional competition shapes proteotoxic ER stress resolution. *Nature Plants* **8**(5):481-490.
 - Highlight in [MSU-PRL Bulletin](#)
12. **Ko DK**, Brandizzi F (2022). Advanced genomics identifies growth effectors for proteotoxic ER stress recovery in *Arabidopsis thaliana*. *Communications Biology* **5**(1):16.
 - Highlight in [MSU-PRL Bulletin](#)
 - Author interview [video](#)
13. York LM, Cumming JR, Trusiak A **Ko DK** Yang WH (2022). Bioenergy Belowground: challenges and opportunities for phenotyping roots and the microbiome for sustainable bioenergy crop production. *Plant Phenome Journal* **5**(1):e20028.
14. Plant Cell Atlas Consortium, Ghosh Jha S, Borowsky AT **Ko DK** ... Rhee SY (2021). Vision, challenges and opportunities for a Plant Cell Atlas. *eLife* **10**:e66877.

15. Angelos E, **Ko DK**, Zemelis-Durfee S, Brandizzi F (2021). Relevance of the Unfolded Protein Response to Spaceflight-Induced Transcriptional Reprogramming in Arabidopsis. *Astrobiology* **21**(3):367-380.
16. **Ko DK**, Brandizzi F (2021). A temporal hierarchy underpins the transcription factor-DNA interactome of the maize UPR. *The Plant Journal* **105**(1):254-270.
– Highlight in [MSU-PRL Bulletin](#)
17. **Ko DK**, Brandizzi F (2020). Network-based approaches for understanding gene regulation and function in plants. *The Plant Journal* **104**(2):302-317.
– Featured in [the Society for Experimental Biology's Spring Bulletin](#)
18. Rice S, Fryer E, Ghosh Jha S...The Plant Cell Atlas Consortium (including **Ko DK**) (2020). First plant cell atlas workshop report. *Plant Direct* **4**(10):e00271.
19. Pastor-Cantizano N, **Ko DK**, Angelos E, Pu Y, Brandizzi F (2019). Functional diversification of ER stress responses in Arabidopsis. *Trends in Biochemical Sciences* **45**(2):123-136.
20. Nadakuduti SS, Starker CG, **Ko DK**, Jayakody TB, Buell CR, Voytas DF, Douches DS (2019). Evaluation of Methods to Assess in vivo Activity of Engineered Genome-Editing Nucleases in Protoplasts. *Frontiers in Plant Science* **10**:110.
21. **Ko DK**, Nadakuduti SS, Douches DS, Buell CR (2018). Transcriptome profiling of transgenic potato plants provides insights into variability caused by plant transformation. *PLoS One* **13**(11):e0206055.
22. **Ko DK***, Rohozinski D*, Song Q, Taylor SH, Juenger TE, Harmon FG, Chen ZJ* (2016). Temporal shift of circadian-mediated gene expression and carbon fixation contributes to biomass heterosis in maize hybrids. *PLoS Genetics* **12**(7):e1006197.
*These authors contributed equally.
23. Shi X, Zhang C, **Ko DK**, Chen ZJ (2015). Genome-wide dosage-dependent and -independent regulation contributes to gene expression and evolutionary novelty in plant polyploids. *Molecular Biology and Evolution* **32**(9):2351-2366.
24. **Ko DK**, Lee MO, Hahn JS, Kim BG, Hong CB (2009). Submergence-inducible and circadian rhythmic basic helix-loop-helix protein gene in *Nicotiana tabacum*. *Journal of Plant Physiology* **166**(10):1090-1100.

Book Chapters and Editorials

1. **Ko DK**, Brandizzi F (2023). Multi-omics resources for understanding gene regulation in response to ER stress in plants. In: Kriechbaumer V (ed.), *The Plant Endoplasmic Reticulum. Methods in Molecular Biology*, vol. 2772. Humana, New York, NY.
2. **Ko DK[#]**, Sanchez-Ballesta MT (2023). Editorial: methods, applications, and protocols in plant science: network modeling-guided understanding of gene regulation in plants. *Frontiers in Plant Science* **14**:1171846.
[#]Corresponding author.
3. **Ko DK**, Brandizzi F (2022). Coexpression network construction and visualization from transcriptomes underlying ER stress responses. In: Lois LM, Trujillo M (eds.), *Plant Proteostasis. Methods in Molecular Biology*, vol. 2581. Humana, New York, NY.

In Revision / Under Review

1. Danieli R*, **Ko DK***, Thomson J, Ruberti C, Brandizzi F. miRNA processing and UPR signaling converge on ER stress recovery through the bZIP28–SAID1 axis. *In revision (Nature Plants)*.
*These authors contributed equally.
2. Kim JK, **Ko DK**, Brandizzi F. The MAP kinase scaffold MORG1 shapes cell death in unresolved ER stress in Arabidopsis. *In revision (Nature Communications; bioRxiv, 2025.01.08.632046)*.
3. **Ko DK[#]**, Thibivilliers S, Libault M[#], Brandizzi F[#]. A DAP-seq atlas of abiotic stress-responsive transcription factor networks in sorghum. *In revision*.
[#]Co-corresponding author.

4. **Ko DK**, Kim SJ, Brandizzi F. Predictive modeling reveals context-dependent transferability of CSLF6-induced transcriptional responses between *Brachypodium distachyon* and *Sorghum bicolor*. *Under review*.

Research Support and Competitive User-Facility Awards

DOE-JGI Bioenergy Research Center user project (Project ID: 1606837) 2025

- Project title: *Sorghum bicolor* Btx623 Abiotic Stress 2.0 Gene Expression Profiling
- Description: Generated time-resolved RNA-seq datasets from sorghum roots and shoots under abiotic stress to identify master transcription factors and regulatory modules underlying stress-responsive gene expression.
- Role: Principal Investigator

DOE-JGI Bioenergy Research Center user project (Project ID: 1437747) 2023

- Project title: *The DNA Binding Landscape of Sorghum TFs in Response to Abiotic Stress*
- Description: Profiled DNA-binding sites of 142 abiotic stress-responsive sorghum transcription factors using DNA Affinity Purification sequencing (DAP-seq).
- Role: Principal Investigator

MSU Project GREEN 2022–2023

- Project title: *Assessing the Impact of Abiotic Stress on Activities of Gene Regulatory DNA Sequences in Sorghum at Single-cell Resolution*
- Role: Principal Investigator
- Amount: \$30,000

DOE-JGI Bioenergy Research Center user project (Project ID: 1331462) 2021

- Project title: *Sorghum bicolor* Btx623 Abiotic Stress Gene Expression Profiling
- Description: Supported RNA-seq library construction and next-generation sequencing of 104 sorghum samples under abiotic stress conditions.
- Role: Principal Investigator

Honors, Awards & Fellowships

Honors & Awards

- GLBRC Yaoping Zhang Bioenergy Research Award 2026
 - Recognizes high-impact GLBRC research contributions advancing sustainable lignocellulosic biofuels and bioproducts.
 - Award [profile](#)
 - News [Highlight](#)
- ASPB Travel Award 2025
- Second Place, Best Poster Award, 2nd Plant Cell Atlas Symposium 2022
- Nominee, Teaching Assistant Award, The University of Texas at Austin 2013
- Great Teaching Assistant Honor, School of Biological Sciences, Seoul National University 2007

Fellowships

- MSU Cloud Computing Fellow 2020-2021
 - Description: Supported cloud computing infrastructure through Microsoft Azure for research projects and participated in hands-on training and public engagement activities.
 - Public [Announcement](#)
- MSU-PRI Travel Grant 2020
- NSF Plant Cell Dynamics Travel Grant 2019
- Bennett Memorial Graduate Fellowship, UT-Austin 2015
- Graduate School Travel Award, UT-Austin 2014

- Summer Research Fellowship, UT-Austin 2013–2014
- Pre-emptive Fellowship, UT-Austin (full scholarship) 2009–2010

Service & Outreach

Chair, PRL Seminar Committee 2026–Present

GLBRC Ambassador 2024

- Disseminating GLBRC's research and promoting bioenergy careers to students from diverse backgrounds
- Attended the Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS) conference in Phoenix, AZ (Oct 31–Nov 2, 2024)

Seminar Host, MSU and UT-Austin

- Speaker: Dr. Samuel Leiboff (Oregon State University) Oct 2024
- Speaker: Dr. Vladimir Gligorijevic (Flatiron Institute) Dec 2019
- Speaker: Dr. Zachary B. Lippman (Cold Spring Harbor Laboratory) Apr 2012

NASA Open Science Data Repository Workshop

- Invited participant, Washington, DC Nov 2023

Core Network Member of the [Plant Cell Atlas](#) 2021–Present

Organizer for Crop Engineering Community of Practice in GLBRC 2023–2024

- Organized monthly webinars for the GLBRC community
- Invited speakers and moderated discussions

GLBRC Annual Science Meeting Committees

- 2024 Planning Committee 2023–2024
- 2023 Session Organizing Committee 2022–2023
- 2022 Session Organizing Committee 2021–2022

DOE-BRC Workshop on AI-ML for Biosystems Design

- Invited participant, Washington, DC Feb 2020

Public Outreach Events

- Organized Grade-1 Plant Day, MSU, East Lansing, MI Sep 2023
- Participated in Family Day Weekend, UT-Austin, Austin, TX Oct 2012

Panel Reviewer

- JGI panel reviewer: Community Science Program (CSP) Functional Genomics 2024
- DOE panel reviewer: DOE SBIR/STTR 2020

Journal Peer Reviewer

- Ad hoc reviewer for approximately 12 manuscripts annually.
- Selected journals include Nature Communications, PNAS, Science Advances, Developmental Cell, The Plant Cell, New Phytologist, and The Plant Journal.

Topic Editor

- *Frontiers in Plant Science* ([Topic page](#))

Teaching

Teaching Assistant, The University of Texas at Austin

Responsibilities across lecture courses included leading weekly discussion sections, delivering concise concept summaries, evaluating examinations and written assignments, supporting student presentations, and holding office hours.

- **BIO327 Genomics** Spring 2016
 - Graduate lecture course; 30 students

- Instructor: Dr. Z. Jeffrey Chen
- **BIO395 Genetics** Spring 2014
 - Upper-division undergraduate lecture course; 23 students
 - Instructors: Drs. Jeffrey Gross & Z. Jeffrey Chen
- **BIO325 Genetics** Fall 2010, Spring/Summer 2011, Spring/Summer/Fall 2012, Spring/Fall 2013
 - Sophomore-level lecture course; 35–121 students per semester
 - Instructors: Drs. Inder M. Saxena, Sibum Sung, and Beverly Finklea
- **BIO205L Laboratory Experiments in Cell and Molecular Biology** Spring 2011
 - Introductory laboratory course; 29 students
 - Instructor: Dr. A. William Allen
 - Led weekly laboratory sessions, introduced experimental concepts, demonstrated techniques, contributed to laboratory design, and evaluated reports, examinations, and term papers.

Teaching Assistant, Seoul National University

- **010.323-003 General Biology Experiment** Fall 2007
 - Introductory laboratory course; 52 students
 - Led weekly laboratory sessions, introduced experimental concepts, demonstrated techniques, and evaluated reports, examinations, and term papers.

Guest Lectures and Workshops

- **Gene Network Workshop**, Chung-Ang University, South Korea. Virtual. Apr 2023
- **Coexpression Network Workshop**, Chung-Ang University, South Korea. Virtual. Oct 2022
- **ChIP-seq Workshop**, Chung-Ang University, South Korea. Virtual. Apr 2022
- **PLB801 Foundation of Plant Biology**, Michigan State University, Virtual. Oct 2020
 - Instructor: Dr. Jiming Jiang
- **BIO325 Genetics**, The University of Texas at Austin Oct 2012
 - Instructor: Dr. Inder M. Saxena

Teaching interests: Space life sciences; plant molecular and stress biology; genetics and genomics; regulatory genomics and bioinformatics; systems biology; and quantitative plant biology.

Mentoring & Trainee Development

Mentoring activities

- Provided guidance on research design, data analysis, scientific writing, presentation preparation, and career development.
- Mentored undergraduate students and early-career researchers in genomics, bioinformatics, plant molecular biology, and scientific communication.
- Supported trainees in developing clear research narratives for presentations, posters, and professional development opportunities.
- Helped trainees understand the broader significance of their research and communicate their work to specialized and interdisciplinary audiences.

Selected mentees

- Marie Brockhoff, GLBRC Research Technician 2026–Present
- Samantha Waite, GLBRC Research Technician 2025–2026
- Ashlyn Savoie, GLBRC Research Technician 2024–2025
- Zachary Smith, GLBRC Research Technician 2023–2024
- Joshua Deradoorian, GLBRC Research Technician 2022–2023
- Chloe Hollidays, MSU Undergraduate Student 2022–2023

- Rita E. Barr, MSU Undergraduate Student 2021–2022
- Sara Knapp, GLBRC Research Assistant 2020–2021
- Elizabeth Selby, MSU Research Assistant 2019–2020
- Rebecca Selby, MSU Research Assistant 2018–2019
- Michael Cadell, UT-Austin Undergraduate Student 2012–2014

Summer undergraduate research mentees

- Xiaohu (Sherry) Sun, GLBRC Summer Undergraduate Research Program May–Jul 2022
 - [Online blog](#) describing her experiences with my mentorship
- Krystal Chi-Shuan Chang, UT-Austin NSF REU May–Jun 2014

Seminar Presentations

Invited Seminars

- GLBRC Crop Engineering Community of Practice. Virtual. Apr 2025
 - Title: *Decoding the Abiotic Stress-Inducible Cistrome of Sorghum*
- Department of New Biology, DGIST, Daegu, South Korea. Virtual. Dec 2024
 - Title: *Decoding Plant UPR Gene Regulation: Insights from Genetics, Genomics, and Network Analysis*
- Biological Research Information Center (BRIC) Webinar Series. South Korea. Virtual. Oct 2023
 - Title: *Unleashing Cellular Destiny: IRE1-Proteasome Signaling Governs Plant ER Proteotoxic Stress*
- GLBRC Bioinformatics Community of Practice. Virtual. Sep 2023
 - Title: *Decoding the Abiotic Stress-Inducible Cistrome of Sorghum*
- Department of Horticulture, Chung-Ang University, South Korea. Virtual. Jul 2021
 - Title: *Network-Enabled Understanding of Gene Regulation in Response to Abiotic Stress*
- Department of Horticulture, Chung-Ang University, South Korea. Jul 2019
 - Title: *Systems-Level Approach to Understand UPR Gene Regulation in Arabidopsis*
- Systems/Synthetic Agrobiotech Center, Gyeongsang National University, South Korea. Jul 2019
 - Title: *Systems-Level Approach to Understand UPR Gene Regulation in Arabidopsis*
- Monsanto Company, Chesterfield, MO Aug 2017
 - Title: *Genome Editing Methods in Potato*

Contributed Seminars

- Center for Sorghum Improvement 2024 Seminar Series. Virtual. Mar 2024
 - Title: *Network-Enhanced Gene Discovery Pipeline (NEEDLE) for Non-Model Plant Species*
- NASA GeneLab Plants AWG meeting. Virtual. Nov 2023
 - Title: *Network-Enhanced Gene Discovery Pipeline (NEEDLE) for Non-Model Plant Species*
- Tuesday Noon Seminar, MSU-DOE Plant Research Laboratories, East Lansing, MI. Feb 2020
 - Title: *A Temporal Hierarchy Underpins the Transcription Factor-DNA Interactome of the Maize UPR*
- Plant Resilience Brown Bag, MSU, East Lansing, MI. Virtual. Dec 2019
 - Title: *A Systems-Level Approach to Understand Transcriptional ER Stress Response in Plants*
- Tuesday Noon Seminar, MSU-DOE Plant Research Laboratories, East Lansing, MI. Apr 2019
 - Title: *Systems-Level Approach to Discover Architecture and Dynamics of ER Stress Gene Networks*
- Plant Luncheon Seminar Series, UT-Austin Mar 2014

- Title: *Early Establishment of Growth Vigor in Maize Hybrids by Circadian Regulators*

Conference Presentations

Invited Talks

- Plant and Animal Genome Conference 34 (PAG 34), San Diego, CA. Jan 2027 (scheduled)
 - Brachypodium Workshop
 - Title: *From Perturbation to Prediction: Modeling CSLF6 Transcriptomic Responses in Brachypodium and Sorghum*
- GLBRC Annual Science Meeting, Geneva, WI. May 2026
 - Title: *A DAP-seq atlas of 142 abiotic stress-responsive sorghum TFs with single-cell multiome insights*
- GLBRC Annual Science Meeting, Milwaukee, WI. May 2025
 - Title: *Mapping the abiotic stress-responsive cistrome in sorghum*
- KSEA Midwest Regional Conference, Urbana-Champaign, IL. Mar 2025
 - Title: *Decoding abiotic stress resilience in sorghum: A transcriptomic framework for climate-ready crops*
- GLBRC Annual Science Meeting, Geneva, WI. May 2023
 - Title: *Network-Enabled Gene Discovery Pipeline (NEEDLE) for Bioenergy Research*
- Biotechnology and Environmental Technology Symposium, South Korea. Virtual. Dec 2022
 - Title: *Network-Enabled Regulatory Dissection of the Mixed-Linkage Glucan Synthase Genes in Grasses*
- Conference for the Recent Advances in New Agro Food Research. Virtual. Jul 2022
 - Title: *Transcriptional Competition Shapes Proteotoxic ER Stress Resolution*
- GLBRC Sustainability Meeting. Virtual. Feb 2022
 - Title: *Network Modeling of Dynamic Transcriptomes Underlying the Development of B. distachyon and S. bicolor*

Selected Contributed Talks

- CSHL meeting - Plant Genomes, Systems Biology & Engineering (Flash talk) Dec 2025
 - Title: *A DAP-seq atlas of abiotic stress-responsive transcription factor networks in sorghum*
- The first annual WALII Symposium: Exploring Life Without Water. Virtual Nov 2025
 - Title: *Decoding Abiotic Stress Resilience in Sorghum: A Transcriptomic Framework for Climate-Ready Crops*
- Phytochemical Society of North America (PSNA) Annual Meeting, East Lansing, MI. Jul 2023
 - Title: *Unleashing Cellular Destiny: IRE1-Proteasome Signaling Governs Plant ER Proteotoxic Stress*
- ASPB Midwest Conference, Ames, IA. Apr 2023
 - Title: *A Signaling Cohort of IRE1 and Proteasome System Controls Cell Fate Determination in Unresolved Proteotoxic Stress of the Plant Endoplasmic Reticulum*
- Plant Cell Dynamics VIII 22nd Plant Biology Symposium, State College, PA. Jun 2019
 - Title: *Gene Networks in the Endoplasmic Reticulum Stress*

Poster Presentations

- CSHL meeting - Plant Genomes, Systems Biology & Engineering Dec 2025
 - Title: *A DAP-seq atlas of abiotic stress-responsive transcription factor networks in sorghum*
- Plant Biology 2025, Milwaukee, WI. Jul 2025
 - Title: *Decoding Abiotic Stress Resilience in Sorghum: A Transcriptomic Framework for Climate-Ready Crops*

- AGBT Agricultural Meeting, Phoenix, AZ. Apr 2024
- Title: *Unraveling Gene Expression Dynamics in Bioenergy Crops Under Environmental Stress Challenges*
- Genomic Science Program Annual Principal Investigator Meeting, Washington, DC Apr 2023
- Title: *Defining Transcriptomic Dynamics in Sorghum in Multiple Abiotic Stresses*
- KSEA Midwest Regional Conference, Chicago, IL. Mar 2023
- Title: *Defining gene expression dynamics in bioenergy crop under environmental stress*
- Plant Cell Atlas Symposium, Virtual Dec 2022
- Title: *Network-Enabled Regulatory Dissection of the Mixed-Linkage Glucan Synthase Genes in Grasses*
- Joint Genome Institute User Meeting, Virtual Aug 2022
- Title: *Network-Enabled Dissection of Conserved and Diverged Regulation of the Mixed-Linkage Glucan Synthase Gene in Bioenergy Crops*
- International Conference on Arabidopsis Research, Virtual. Jun 2022
- Title: *Transcriptional competition shapes proteotoxic ER stress resolution*
- Plant Cell Atlas Symposium, Virtual Dec 2021
- Title: *Understanding Gene Regulatory Networks Underlying Plant ER Stress: Insights from Bulk Cells to Single Cell*
- Plant Biology Worldwide Summit, Virtual. Jul 2021
- Title: *Gene-Regulatory Network-Enabled Identification of Effectors Controlling Organ Growth During Recovery from Endoplasmic Reticulum Stress*
- ICAR 2021, Virtual. Jun 2021
- Title: *Gene-Regulatory Network-Enabled Identification of Effectors Controlling Organ Growth During Recovery from Endoplasmic Reticulum Stress*
- EMBO Workshop - International Plant Systems Biology, Virtual. Apr 2021
- Title: *Gene-Regulatory Network-Enabled Identification of Effectors Controlling Organ Growth During Recovery from Endoplasmic Reticulum Stress*
- BRC Workshop: AI and Machine Learning for Biosystems Design, Washington, DC Feb 2020
- Title: *Systems-Level Analysis of ER Stress Gene Networks in Plants*
- 59th Annual Maize Genetics Conference, Saint Louis, MO. Mar 2017
- Title: *Expansion of the Wisconsin Diversity Panel to Further Document the Maize Pan-Transcriptome*
- CSHL Meeting - Plant Genomes & Biotechnology, Woodbury, NY. Dec 2015
- Title: *Circadian-Mediated Regulation of Morning-Phased Genes Contributes to Biomass Heterosis in Maize Hybrids*
- ASPB 2007, Chicago, IL. Jul 2007
- Title: *Functional Analysis of Heat Shock Protein 70 in Flooding-Stressed *Nicotiana tabacum**

Professional Societies

American Society of Plant Biologists (ASPB), Active Member